Smart Water Quality Monitoring

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Date: 1/3/2026

1. Introduction

Water quality monitoring plays a vital role in public health, environmental protection, and industrial safety. Conventional testing methods rely heavily on manual sampling and laboratory analysis, resulting in delayed insights and limited responsiveness to contamination events. To address these limitations, this project implements an IoT-based Smart Water Quality Monitoring System capable of continuous, real-time observation.

The system integrates multiple sensors with an ESP32 microcontroller to measure critical parameters and transmit data to a cloud platform for remote monitoring, analytics, and anomaly detection.

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2. Problem Statement

Develop an IoT-based water quality monitoring system capable of measuring key parameters such as pH, turbidity, temperature, and dissolved oxygen in real time. The system should display live readings locally and upload data to a cloud platform. The cloud analytics module should detect abnormal readings, generate alerts, and predict potential water quality degradation trends.

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3. System Objectives

The primary objectives of the system are:

- Real-time monitoring of water parameters
- Cloud-based remote visualization
- Detection of unsafe water conditions
- Early identification of contamination risks
- Support for preventive maintenance strategies

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4. System Design & Architecture

The system follows a layered IoT architecture:

Sensor Layer – Captures physical water parameters
Processing Layer – ESP32 handles computation & filtering
Communication Layer – Wi-Fi data transmission
Cloud Layer – Data storage, visualization & analytics

Components Used

- ESP32 Development Board
- pH Sensor
- Temperature Sensor (DS18B20)
- Ultrasonic Sensor (Water Level Monitoring)

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5. Hardware Implementation

5.1 Hardware Setup

The prototype system was assembled using an ESP32 DevKit board interfaced with the sensors. The sensors were arranged to allow stable immersion and accurate measurement.

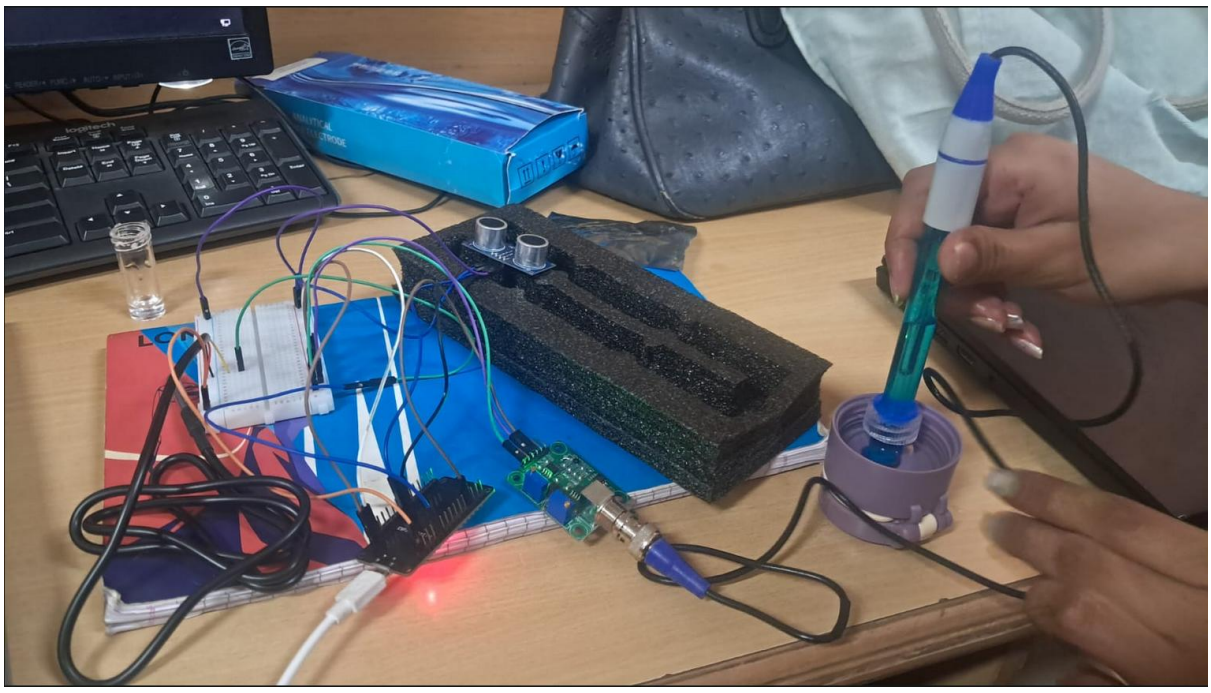

Figure 1 – Experimental Hardware Setup

The ultrasonic sensor was positioned above the water container to measure water level. The pH probe was immersed directly into the sample solution. The temperature sensor was placed within the water body to capture thermal variations.

5.2 ESP32 Microcontroller

The ESP32 serves as the central controller responsible for:

- Reading analog and digital sensor signals
- Applying noise reduction via averaging
- Performing calibration computations
- Uploading data to ThingSpeak via Wi-Fi

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6. Working Principle

The system operates through the following sequence:

1. Sensors capture real-time measurements
2. ESP32 reads sensor outputs
3. Multiple samples averaged for stability
4. Raw signals converted to engineering units
5. Data displayed via Serial Monitor / LCD (if connected)
6. Data transmitted to ThingSpeak Cloud
7. Analytics detect abnormal conditions

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7. Software & Firmware Design

The ESP32 firmware was developed to perform reliable sensing, filtering, and communication.

Key Functional Modules

- Wi-Fi connectivity management
- pH sensor voltage conversion & calibration
- Temperature acquisition (DS18B20 – OneWire)
- Ultrasonic distance measurement
- Moving average noise filtering
- Threshold-based anomaly detection
- Cloud upload using HTTP protocol

Noise Reduction Strategy

To minimize sensor instability, averaging across multiple samples was implemented:

- pH readings averaged over 10 samples

- Ultrasonic distance averaged over 10 samples

This significantly reduced fluctuations caused by electrical noise and environmental disturbances.

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8. Cloud Integration (ThingSpeak)

Sensor data was uploaded to ThingSpeak for visualization and analysis.

Field Mapping:

- Field 1 → pH Sensor
- Field 2 → Temperature Sensor
- Field 3 → Ultrasonic Sensor (Water Level)

ThingSpeak enabled:

- Real-time graphical visualization
- Historical trend monitoring
- Remote accessibility

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9. Contamination Detection Strategy

Contamination risks were inferred using threshold-based evaluation:

- **pH Abnormality:** Outside safe drinking range (6.5 – 8.5)
- **Temperature Anomaly:** Elevated thermal readings
- **Water Level Anomaly:** Sudden drops or overflow conditions

The ESP32 performs local anomaly checks and flags unsafe conditions.

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10. Results & Observations

10.1 Cloud Visualization Outputs

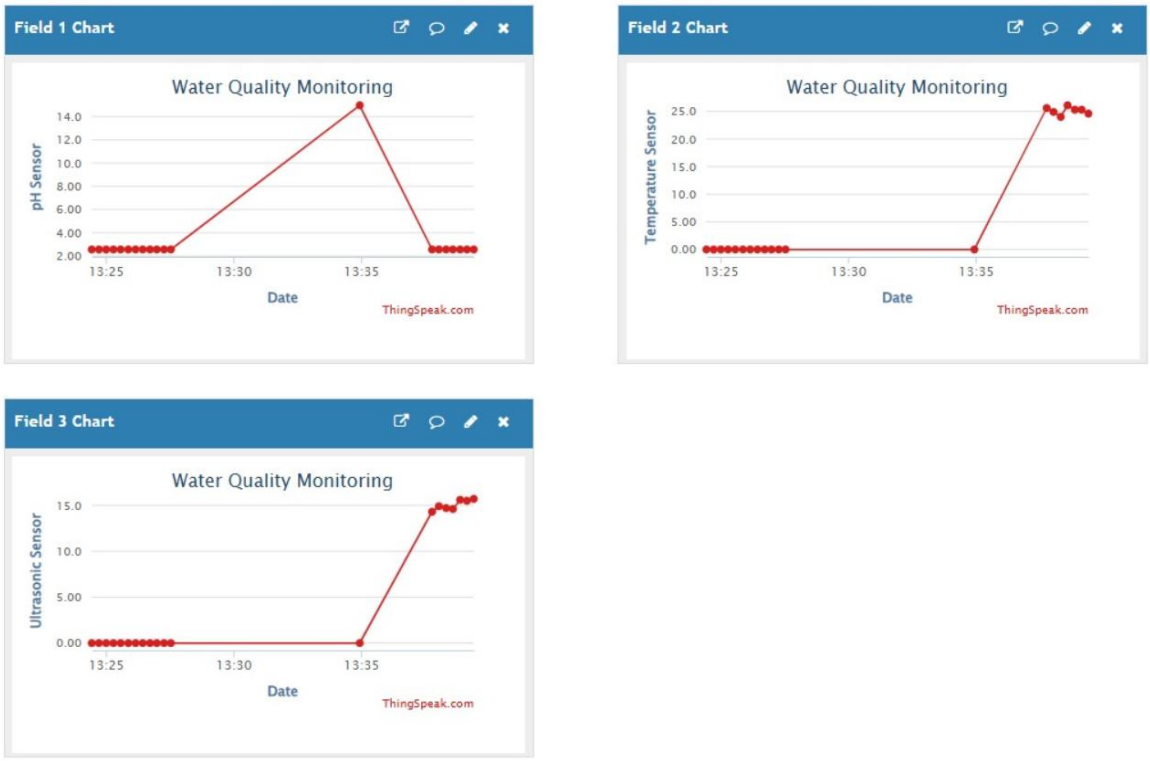

Figure 2 – ThingSpeak Sensor Graphs

10.2 Observed Behaviour

pH Sensor Behaviour:
The graph shows a stable baseline followed by a sudden spike reaching extreme values before returning to nominal levels. This behaviour indicates either:

- Sensor disturbance during probe movement
- Calibration transient response
- Intentional contamination simulation

Such spikes demonstrate the system's ability to capture rapid chemical variations.

Temperature Sensor Behaviour:
Temperature remained near baseline initially, followed by a sharp rise. This may correspond to:

- Sensor immersion event
- Environmental temperature change

- Heat introduction experiment

Ultrasonic Sensor Behaviour:
The water level graph shows a step increase, suggesting:

- Water addition into container
- Sensor repositioning
- Tank level variation simulation

10.3 System Validation

The recorded variations confirm:

- Correct sensor interfacing
- Successful cloud transmission
- Real-time responsiveness of the IoT system

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11. Challenges Encountered

Several practical challenges were observed:

- pH sensor calibration sensitivity
- Analog signal noise
- Ultrasonic echo inconsistencies
- Sensor placement stability

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12. Preventive Recommendations

To enhance system reliability:

- Periodic sensor calibration
- Shielded wiring for analog sensors
- Median filtering for ultrasonic readings
- Stable probe mounting mechanisms

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13. Conclusion

The IoT-based Smart Water Quality Monitoring System successfully demonstrates continuous sensing, real-time cloud visualization, and anomaly detection. The system proves that low-cost

embedded IoT solutions can effectively support environmental monitoring and early contamination awareness.

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14. Future Enhancements

Potential improvements include:

- Integration of turbidity sensor
- Dissolved oxygen measurement
- Machine learning-based contamination prediction
- Mobile dashboard & alerts